

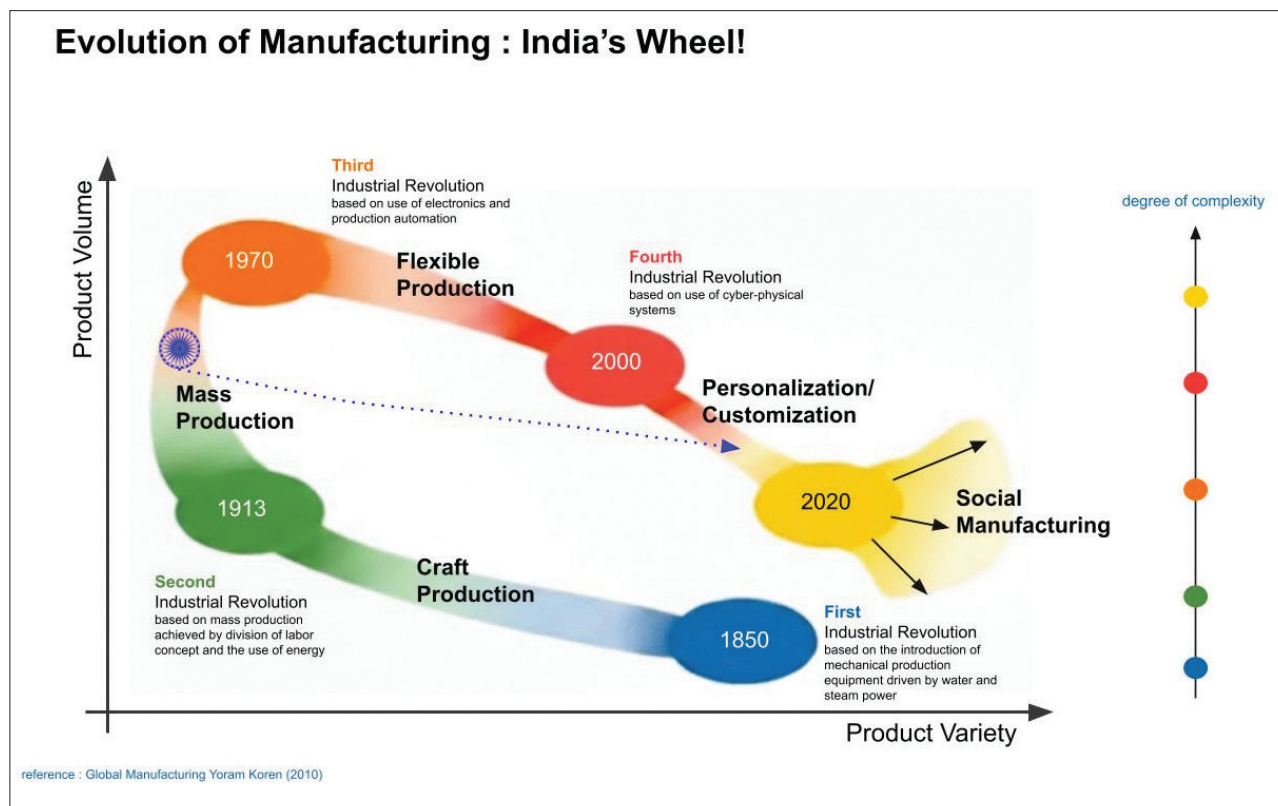


Figure 2: Explaining the evolution of manufacturing at the EFY Expo Gujarat 2025

mass manufacturing. The real challenge is to leapfrog into the next generation of manufacturing.

The common response to this challenge is smart manufacturing. But that immediately raises another question. What do we really mean by smart manufacturing? When we look closely, it appears as a mix of technologies brought together through cross-pollination. The deeper issue, however, is whether these technologies are truly suited to Indian manufacturing realities.

This is where sustainability becomes critical. Most of these technologies are push-driven. They exist because they are available, not because industry is ready to adopt them. And their adoption is constrained by scalability, maturity, and cost. Industries often invest millions in building models but hesitate to invest even a few thousand rupees in building quality data. That imbalance creates friction and limits meaningful impact.

Democratising smart manufacturing for India's fraternal economy

Looking inward at Indian manufacturing, India's economy is largely powered by what I call the fraternal economy, driven by small and medium manufacturing units. These enterprises are not just job creators. They are wealth creators and form the backbone of industrial growth. This segment is also the true force multiplier that can lead India towards a manufacturing renaissance.

A clear disconnect today is that, on the one hand, we have advanced technologies and, on the other, end users who need solutions that are simple, affordable and reliable. Bridging this gap becomes essential for making smart manufacturing meaningful and sustainable in India. Which brings us to the next question: how do we Indianise smart manufacturing technologies? While much has been said about

tools and platforms, the real drivers of empowerment are talent and tenacity. The first step towards unlocking both is the democratisation of technology.

Let us look at a real-world example to understand this better. Consider a small-scale factory operating within the fraternal economy. Such factories often struggle with transaction settlement, traceability, and digital record-keeping. Typically, one individual handles everything from operations to accounting. It is unrealistic to expect this person to embrace sophisticated industrial technologies. We must expand on what already works rather than adding more layers of complexity. Digital payments offer a clear example. Unified payment interface (UPI) is one of India's strongest digital backbones, and it can be extended into manufacturing workflows. In fact, we have done exactly that by integrating UPI with open source technologies to create a simple, sustainable smart solution that addresses real operational needs.